

System Logging

Objectives

After completing this lesson, you should be able to:

- Describe Oracle Linux 7 system logging options
- Describe the contents of the `rsyslog` configuration file
- Describe `rsyslog` filter options
- Describe facility/priority-based filters
- Describe `rsyslog` actions
- Describe `rsyslog` templates
- Configure log rotation
- Describe `logwatch`
- Describe `journald`
- Use the `journalctl` utility
- Explain process accounting



System Logging: Introduction

- Log files store system, kernel, service, and application messages.
 - Most log files are located in the `/var/log/` directory.
- Some log files are controlled by the `rsyslogd` daemon.
 - `/etc/rsyslog.conf` is the main configuration file.
 - It contains global directives, modules, and rules.
- With Oracle Linux, log files can also be managed by the `journald` daemon.
 - `journald` is a component of `systemd`.
 - `journald` captures various system messages, indexes them, and stores them in `/run/log/journal/`.
 - Use the `journalctl` utility to view the journal logs.



rsyslog Configuration

- Global directives:
 - They specify configuration options that apply to the `rsyslogd` daemon.
 - All configuration directives must begin with a dollar sign (\$).

```
$IncludeConfig /etc/rsyslog.d/*.conf
```

- Modules:
 - Are dynamically loaded by using the `$ModLoad` global directive
 - Provide additional functionality and configuration directives
 - Categories of modules include *Input*, *Output*, *Parser*, *Message modification*, *String generator*, and *Library*.
- Rules:
 - Specify a *filter* (`cron.*`) and *action* (log all `cron` messages to `/var/log/cron`):

```
cron.* /var/log/cron
```

rsyslog Filter Options

There are three ways to filter `rsyslog` messages:

- Facility/priority-based filters:
 - Filters are based on two conditions: facility and priority.
 - Facility specifies the subsystem that produces the message.
 - Priority represents the priority of the message.
- Property-based filters:
 - Filter by comparing a property of the message to a value

```
:msg, contains, "error"
```

- Expression-based filters:
 - Filter according to arithmetic, Boolean, or string operations
 - Use `rsyslog` scripting language. Syntax:

```
if EXPRESSION then ACTION else ACTION
```

Facility/Priority-Based Filters

Messages are filtered based on two conditions: Facility and priority.

- Syntax to create a filter (or selector):

Facility.Priority

- Select all `auth rsyslog` messages with any priority:

`auth.*`

- Select all `mail rsyslog` messages with priority `err` and higher:

`mail.err`

- Select all `user rsyslog` messages except those with `info` or `debug` priority:

`user.!info,!debug`

rsyslog Actions

- Actions specify what to do with the filtered messages.
- Options include:
 - Save `rsyslog` messages to log files.
 - Send `rsyslog` messages over the network.
 - Send `rsyslog` messages to specific users.
 - Execute a program.
 - Write `rsyslog` messages into a database.
 - Discard `rsyslog` messages.
- To save `cron` messages to `/var/log/cron.log`:

```
cron.* /var/log/cron.log
```

- To send `rsyslog` messages over the network:

```
*.* @example.com:18
```

rsyslog Templates

Templates modify and format output generated by `rsyslog`.

- Syntax to create a template:

```
$template TEMPLATE_NAME, "text %PROPERTY% text", [OPTION]
```

- Templates can be used to generate dynamic file names:

```
$template DynamicFile, "/var/log/%timegenerated%-test.log"
```

- Example of a template definition:

```
$template class, "Time: %timestamp%, Severity: %syslogfacility-text% -  
%syslogpriority%, Program: %programname% [%procid%] %msg%\n"
```

- Example of using a template in a rule:

```
*.* $DynamicFile;class
```


Configuring Log Rotation (`logrotate`)

- `logrotate` is a utility to automatically manage log files.
 - It runs as a daily `cron` job using the `/etc/cron.daily/logrotate` file.
- The `/etc/logrotate.conf` file is the global configuration file for all logs.
 - The `/etc/logrotate.d/` directory contains a separate configuration file for any specific log file.
- Configuration options include:
 - How often to rotate files
 - The number of rotated log files to keep
 - Scripts to run before or after rotating
 - Specify log files to be mailed
 - Enable compression of log files
- See `man logrotate` for a list of options.

The options in the `/etc/logrotate.d/chrony` configuration file are described as follows:

- **missingok:** If the log file is missing, do not issue an error message.
- **nocreate:** New log files are not created.
- **postrotate/endscript:** The lines between these directives are executed after the log file is rotated.
- **sharedscripts:** The `postrotate` script runs only once, not once for each log that is rotated.

The logwatch Utility

- logwatch is a utility to perform basic log file monitoring and analysis.
 - It runs as a daily cron job by using the `/etc/cron.daily/0logwatch` file.
- The local configuration file is:
 - `/etc/logwatch/conf/logwatch.conf`
- The main configuration file is:
 - `/usr/share/logwatch/default.conf/logwatch.conf`
- A HOWTO-Customize-Logwatch file exists in the following directory:
 - `/usr/share/doc/logwatch-<version>/`
- logwatch can also be run from command line. Example:

```
# logwatch --help
```

journald: Introduction

- Journald:
 - Is a logging service included with `systemd`
 - Collects and stores logging data in structured, indexed journals.
 - Supports advanced query options and faster search times
 - Adds structured metadata to the messages that assist in troubleshooting
 - Can be used together with, or in place of, `rsyslogd`
- Journals are nonpersistent by default.
 - Are stored in `/run/log/journal/`
- To configure persistent journal data storage:

```
# mkdir -p /var/log/journal
# systemctl restart systemd-journald
```

The journalctl Utility

- Use the `journalctl` command to view the journal logs.

```
# journalctl
```

- Output is displayed one page at a time.
- Time stamps are converted to your local time zone.
- Priority of entries is visibly marked.
 - Entries with error priority and higher are red.
 - Entries with notice and warning priority are in bold font.
- The beginning of the boot process has a special entry.
- With no options, all log data is displayed.
- By default, oldest entries are listed first.
- A number of query options are available. See:

```
# journalctl -h
```

Journalctl - advanced filtering (1 of 3)

Show full content with oldest entries

```
# journalctl
```

Show full content with details

```
# journalctl -o verbose
```

Show all system related and jump to end of journal

```
# journalctl --system -e
```

Additional warning / errors with possible solutions

```
# journalctl -xe
```

Show additional msg and perform tail like

```
# journalctl -xef
```

Journalctl - advanced filtering (2 of 3)

Filter message based on date & time

```
# journalctl -S "2020-09-29 13:00:00"  
# journalctl --since "2020-11-20"  
# journalctl --since today  
# journalctl -S "2020-01-01 09:00" -U "2021-01-01 17:00"  
# journalctl -S "-30 minute"  
# journalctl -S "-1 hour"  
# journalctl -S "-1 day"  
# journalctl -S "-1 week"
```

Journalctl - advanced filtering (3 of 3)

Retrieves logs limited to current boot session

```
# journalctl -b
```

Retrieves logs limited to second , third boot

```
# journalctl -b 2
```

```
# journalctl -b 3
```

Show all messages rated as error or worse from last boot

```
# journalctl -b -1 -p err
```

Granular search query

```
# journalctl _SYSTEMD_UNIT="crond.service"
```

```
# journalctl _PID=1
```

```
# journalctl _SYSTEMD_UNIT="sshd.service" _PID=1182
```

```
# journalctl -S "9:00" _SYSTEMD_UNIT="crond.service"
```


Quiz



Which of the following entries in `/etc/rsyslog.conf` **cause** warning, err, crit, alert, and emerg messages from the kernel to be logged?

- a. `kern.*`
- b. `kern.warning`
- c. `kern.err`
- d. `*.kern`



Quiz



Which of the following entries in `/etc/rsyslog.conf` **cause** warning, err, crit, alert, and emerg messages from the kernel to be logged?

- a. `kern.*`
- b. `kern.warning`
- c. `kern.err`
- d. `*.kern`



Quiz



Which of the following statements are true?

- a. To use the `journal` service, you must stop `rsyslogd`.
- b. Journals are nonpersistent by default.
- c. The `journal` service adds structured metadata to the messages that assist in troubleshooting.
- d. Use the `journal` command to view the journal log files.



Quiz



Which of the following statements are true?

- a. To use the `journal` service, you must stop `rsyslogd`.
- b. Journals are nonpersistent by default.
- c. The `journal` service adds structured metadata to the messages that assist in troubleshooting.
- d. Use the `journal` command to view the journal log files.



Summary

In this lesson, you should have learned how to:

- Describe Oracle Linux 7 system logging options
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Practices for Lesson 19: Overview

The practices for this lesson cover the following topics:

- Configuring system logging
- Using `rsyslog` templates
- Using `logwatch`
- Using `journald`
- Using process accounting